2. 8-Queen Problem

Aim: To write a Python program to place 8 queens on a chessboard so that no two queens attack each other.

Algorithm:

1. Initialize an empty 8 × 8 chessboard.
2. Place a queen in the first row.
3. Check whether the position is safe.
4. If safe, place a queen in the next row.
5. If no safe position is available, backtrack.
6. Repeat until all 8 queens are placed.
7. Display the chessboard.

Code:

```
def solve (board, row):
    if row == 8:
        for r in range(8):
            print (" “. Join (
                "Q" if board[r] == c else "."
                for c in range (8)
            ))
        return True
    for col in range (8):
        safe = True
        for r in range(row):
            if (board[r] == col or
                abs(board[r] - col) == abs (r - row)):
                safe = False
                break
        if safe:
            board[row] = col
```

```
            if solve (board, row + 1):
                return True
    return False
board = [-1] * 8
solve (board, 0)
```

Input:

8 × 8 chessboard

Output:

```
====================
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

Result: Thus, the 8-Queen Problem was solved successfully using Python.